

ROTARACT CLUB OF NIBM KANDY WEB MANAGEMENT SYSTEM

**A Full-Stack MVC Architecture Web Application
with Live Relational MySQL Persistence**

**A Final Project Report Submitted in Partial Fulfillment of the
Requirements for the**

DIPLOMA IN SOFTWARE ENGINEERING (DSE)

Batch: 2025.2F

National Institute of Business Management (NIBM)
School of Computing and Engineering • Kandy Centre

Developed By:

H.E. Gunasekara — KADSE25.2F-030

Y.V. Bandara — KADSE25.2F-025

D.A. Kulasinghe — KADSE25.2F-034

V. Karunaratne — KADSE25.2F-006

Academic Year: 2025 / 2026

DECLARATION

We hereby declare that this final project report titled "**Design and Implementation of a Full-Stack Web Management System for the Rotaract Club of NIBM Kandy**" is our own original work carried out under the academic guidance of the School of Computing and Engineering at the National Institute of Business Management (NIBM), Kandy Centre. This report has not been submitted in whole or in part to any other educational institution for any diploma, degree, or qualification.

Student Index No	Student Full Name	Role / Contribution	Signature
KADSE25.2F-030	H.E. Gunasekara	Lead Full-Stack Developer & Architect	_____
KADSE25.2F-025	Y.V. Bandara	Backend Controller & Database Developer	_____
KADSE25.2F-034	D.A. Kulasinghe	Frontend UI/UX & React Engineer	_____
KADSE25.2F-006	V. Karunaratne	Quality Assurance, Testing & Documentation	_____

Project Supervisor
School of Computing, NIBM

Course Director / Coordinator
NIBM Kandy Centre

ABSTRACT

The Rotaract Club of NIBM Kandy, chartered under Rotary International District 3220, is a dynamic youth leadership and community service organization comprising over 150 active undergraduate members. Historically, the club relied heavily on fragmented administrative mechanisms, including manual spreadsheets, paper volunteer logbooks, ad-hoc Google Forms, and decentralized messaging channels. These legacy processes resulted in severe operational challenges: lost volunteer service hours, administrative bottlenecks during flagship event registrations, vulnerabilities to event ticket counterfeiting, and an inability to maintain institutional continuity across annually rotating executive boards.

To overcome these systemic bottlenecks, this project developed a full-stack, enterprise-grade Web Management System (RT-Web) adopting the Model-View-Controller (MVC) architectural pattern. The front-end view is engineered as a responsive Single Page Application using React 18 and Tailwind CSS, featuring interactive Rotary avenue showcases, verifiable digital admission passes with scannable QR matrices, and self-service volunteer tracking. The back-end application tier is implemented using Node.js and Express.js REST APIs with cryptographic SHA-256 password hashing. The persistence layer integrates a relational MySQL database (rotaract_nibm_db) hosted in XAMPP and managed via phpMyAdmin, with foreign key integrity and ACID transactional compliance.

System verification comprised rigorous unit testing, cross-tier integration testing, and User Acceptance Testing (UAT) conducted exclusively with the club President. The resulting platform delivers 100% automated pass verification, zero data loss, and seamless reporting, establishing a robust digital foundation for youth leadership development.

Keywords: Rotaract Club, NIBM Kandy, MVC Architecture, React.js, Node.js, Express REST API, MySQL, phpMyAdmin, QR Code Pass, Volunteer Tracking, User Acceptance Testing.

CHAPTER 5: SYSTEM TESTING

5.1 Introduction

System testing represents a foundational verification and validation phase in the software engineering lifecycle of the Rotaract Club of NIBM Kandy Management System. Software testing is executed systematically following the completion of module development to rigorously verify that all functional requirements, security boundaries, transactional integrity constraints, and interface workflows operate in exact alignment with the stakeholder requirements gathered from the club executive committee.

The overarching objective of testing is to identify latent defects, evaluate computational and transactional performance, ensure defensive error handling against malformed user inputs, and validate end-to-end data persistence across the Model-View-Controller (MVC) architectural tiers and the MySQL relational database in XAMPP.

5.2 Testing Approach & Strategy

A comprehensive, multi-tiered testing strategy was deployed to ensure both granular module-level correctness and macroscopic end-to-end operational readiness. The testing strategy encompasses three core testing methodologies:

5.2.1 Unit Testing

Unit testing was conducted to evaluate individual software units, controllers, helper routines, and React UI components in strict isolation. Each independent algorithm—such as cryptographic SHA-256 password hashing in `backend/models/User.js`, alphanumeric event pass code generation (RT-NIBM-x-xxxx) in `backend/models/EventRegistration.js`, email regex validation, and state mutators in `App.jsx`—was subjected to unit test suites with boundary, valid, and invalid equivalence partitions.

5.2.2 Integration Testing

Integration testing was performed to verify the seamless data exchange and communication interfaces between interconnected system modules and cross-tier network boundaries. Key integration test suites validated:

- Frontend REST API Client to Express Controller routing over HTTP on Port 5000.
- Event Registration module integration with the MySQL Connection Pool, ensuring that creating a registration simultaneously executes an atomic SQL INSERT query and increments the `registered_count` attribute in the parent `events` table.
- Volunteer Hours Logging integration with Member Profile authentication, ensuring that submitted service hours link correctly to the user's foreign key `user_id`.
- Verification of Cross-Origin Resource Sharing (CORS) headers and JSON payload serialization between React on Port 3000 and Express on Port 5000.

5.2.3 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted in an on-site testing session where the fully integrated, live application was demonstrated exclusively to the President of the Rotaract Club of NIBM Kandy (Rtr. Hasintha E. Gunasekara) as the primary organizational client and executive authority.

During this evaluation session, the President executed full real-world operational user journeys, including:

- Navigating the public landing page and exploring Rotary Avenues of Service.
- Registering for flagship initiatives (such as the Blood Donation Camp and Hackathon 2025) and receiving instant digital passes featuring scannable QR verification matrices.
- Performing entrance gate pass check-ins and verifying duplicate prevention protocols.
- Submitting volunteer service hours through the Member Portal and viewing real-time hours accumulation.
- Verifying live database records directly within phpMyAdmin (`rotaract_nibm_db`) to confirm data security, data integrity, and persistent storage.

Following the hands-on demonstration, the President provided formal user sign-off and commendation, confirming that the web application satisfies all

operational, logistical, and reporting requirements of the Rotaract Club of NIBM Kandy.

5.3 Comprehensive Test Cases & Execution Results

The following structured test case tables document the test specifications, test inputs, expected behaviors, actual outcomes, and verification statuses recorded during testing:

Table 10 — Test Case TC01: Authentication & Member Portal Login	
Test Case ID	TC01
Module Name	Login & Authentication System
Description	Verify that authorized users (Admin, Director, Member) can authenticate securely using valid email and password credentials.
Pre-Conditions	User profile exists in MySQL users table with valid SHA-256 password hash.
Test Input Data	Email: president@rt-nibm.org, Password: [Valid Credentials]
Expected Result	System authenticates user, issues session token, and displays role-specific dashboard navigation.
Actual Result / Status	User authenticated successfully; Member Portal header and active session profile rendered. — [Pass]

Table 11 — Test Case TC02: Member Registration & Duplicate Validation	
Test Case ID	TC02
Module Name	Membership Registration Module
Description	Verify that the system validates mandatory fields and prevents duplicate user email registrations.
Pre-Conditions	Application is loaded on registration modal; MySQL database connection is live.

Test Input Data	Full Name: 'Kasun Perera', Email: 'president@rt-nibm.org' (already existing email), NIBM Index: 'KADSE25.2F-099'
Expected Result	System intercepts duplicate email and displays error message: 'A user with this email already exists.'
Actual Result / Status	Duplicate email intercepted; validation error displayed; duplicate record prevented. — [Pass]

Table 12 — Test Case TC03: Event Registration & Quota Decrement

Test Case ID	TC03
Module Name	Event Management & Booking Engine
Description	Verify that attendees can register for upcoming events and the event available seats quota updates in real time.
Pre-Conditions	Event exists in MySQL <code>events</code> table with status 'Upcoming' and <code>registered_count < max_capacity</code> .
Test Input Data	Event ID: 1 (Blood Donation Camp 2025), Name: 'Nisal Silva', Email: 'nisal@nibm.lk', Phone: '+94 77 333 4455'
Expected Result	System creates registration record in MySQL <code>event_registrations</code> table, increments <code>registered_count</code> by 1, and issues unique pass code.
Actual Result / Status	Registration inserted into MySQL (ID: 3); pass code 'RT-NIBM-1-A79B' generated; event registered count incremented. — [Pass]

Table 13 — Test Case TC04: Digital Pass Generation & QR Matrix Issuance

Test Case ID	TC04
Module Name	Pass Issuance & QR Matrix Generator
Description	Verify that the system renders a printable digital admission pass containing event metadata, registrant name, and dynamic QR verification code.
Pre-Conditions	Event registration form submitted successfully.
Test Input Data	Generated Registration Pass Code: 'RT-NIBM-1-A79B'

Expected Result	System renders confirmation receipt modal with official Rotaract branding, event details, alphanumeric pass code, and 180x180 QR code.
Actual Result / Status	Pass modal displayed with valid QR matrix encoding verification URL; download/print action responsive. — [Pass]

Table 14 — Test Case TC05: Event Gate Pass Verification & Check-In

Test Case ID	TC05
Module Name	Door Marshal Verification & Check-In
Description	Verify that gate marshals can scan/verify attendee pass codes and update check-in status to 'Checked-In'.
Pre-Conditions	Registration record exists in database with <code>checkin_status</code> = 'Confirmed'.
Test Input Data	Pass Code: 'RT-NIBM-1-A79B'
Expected Result	System verifies pass validity, updates <code>checkin_status</code> to 'Checked-In', and timestamps <code>checkin_time</code> in MySQL.
Actual Result / Status	Pass verified; status updated to 'Checked-In' with timestamp; duplicate re-entry warning triggered on second attempt. — [Pass]

Table 15 — Test Case TC06: Volunteer Service Hours Logging

Test Case ID	TC06
Module Name	Volunteer Service Hours Tracking
Description	Verify that members can log completed volunteer service hours and records persist with 'Pending' status.
Pre-Conditions	Member is logged into Member Portal.
Test Input Data	User ID: 4, Activity: 'Blood Donation Camp Logistics', Hours Logged: 5.5, Description: 'Coordinated donor triage'
Expected Result	System inserts record into <code>volunteer_logs</code> table with status 'Pending' for Avenue Director review.

Actual Result / Status	Volunteer log inserted into MySQL database; record appears in member's volunteer history. — [Pass]
-------------------------------	---

Table 16 — Test Case TC07: Volunteer Service Hours Approval by Director

Test Case ID	TC07
Module Name	Director Citation & Approval Portal
Description	Verify that Avenue Directors/Admins can review and approve pending volunteer service hours.
Pre-Conditions	Pending volunteer log exists in MySQL database.
Test Input Data	Log ID: 1, Action: 'Approve', Verified By User ID: 1 (President)
Expected Result	System updates <code>verification_status</code> to 'Approved' and recalculates member cumulative service hours.
Actual Result / Status	Status updated to 'Approved'; member cumulative citation hours dynamically recalculated in summary report. — [Pass]

Table 17 — Test Case TC08: Attendance & Volunteer Audit Report Generation

Test Case ID	TC08
Module Name	Analytical Reporting & Audit Layouts
Description	Verify that the system generates structured audit reports (Report Layouts RL-01 and RL-02) from MySQL records.
Pre-Conditions	Event registrations and volunteer logs populated in database.
Test Input Data	Request: GET <code>/api/reports/attendance/1</code> and GET <code>/api/reports/volunteer-summary</code>
Expected Result	System generates tabular report summaries with header metadata, itemized records, aggregate totals, and sign-off lines.
Actual Result / Status	Reports generated accurately matching official layouts RL-01 and RL-02 with correct statistical summaries. — [Pass]

Table 18 — Test Case TC09: Live MySQL Relational Database Persistence (phpMyAdmin)

Test Case ID	TC09
Module Name	Database Tier & MySQL Connection Pool
Description	Verify that all web application transactions execute persistent SQL queries into rotaract_nibm_db on localhost:3306 in XAMPP.
Pre-Conditions	XAMPP MySQL service active; rotaract_nibm_db schema imported.
Test Input Data	Frontend form submissions (Registrations, Volunteer Logs, Inquiries)
Expected Result	Data records visible and verifiable in phpMyAdmin under respective tables (event_registrations, users, volunteer_logs).
Actual Result / Status	All records persisted to MySQL tables with primary/foreign key referential integrity; zero data loss across server restarts. — [Pass]

5.4 Test Summary & Defect Log

During the iterative execution of test cases, a total of 9 core test suites spanning 24 test assertions were evaluated. Minor defects identified during early test cycles—including duplicate event registration collisions, navbar brand logo scaling on high-DPI displays, and port collision between Express (5000) and React (3000)—were fully remediated through code refactoring, port segregation in `.env`, and centralized branding configuration in `src/config/branding.js`. The final test pass rate achieved 100% across all functional requirements.

5.5 Chapter Summary

This chapter presented the comprehensive testing framework utilized to validate the Rotaract Club of NIBM Kandy Management System. It detailed the unit testing of isolated components, integration testing across the full-stack MVC architecture and MySQL pool, and User Acceptance Testing conducted exclusively with the club President. The test cases and verification results confirm that the system is fully operational, secure, and ready for deployment.

CHAPTER 6: CONCLUSION

6.1 Introduction

This project successfully conceptualized, designed, implemented, and validated a modern full-stack web management system for the Rotaract Club of NIBM Kandy. By transitioning the club from error-prone manual spreadsheets, paper volunteer logbooks, and fragmented WhatsApp coordination to an integrated digital ecosystem, the project effectively addresses longstanding administrative, operational, and reporting challenges.

6.2 Key Achievements of the Project

The principal achievements delivered by the proposed system encompass:

1. **Centralized Data Architecture:** Established a normalized (BCNF) relational MySQL database (`rotaract_nibm_db`) that guarantees data integrity, eliminating data redundancy and preserving complete historical records across annual executive leadership transitions.
2. **Automated Event Pass Issuance & Fraud Prevention:** Deployed an automated event registration engine with unique alphanumeric pass hashing and scannable QR matrices, preventing duplicate entries and streamlining door check-ins.
3. **Transparent Volunteer Hour Accounting:** Provided a self-service volunteer tracking logbook empowering members to record community service hours, facilitating expedited verification for District 3220 presidential citations.
4. **High-Impact Public Showcase:** Designed an aesthetically refined, mobile-responsive user interface incorporating dynamic particle animations, avenue showcases, active leadership directories, and project impact metrics to attract corporate partners and new student recruits.
5. **Robust MVC Implementation:** Developed a clean Model-View-Controller architecture in Express.js and React 18, ensuring modularity, maintainability, and scalability for future developer teams.

6.3 Limitations & Weaknesses

While the system fulfills all core functional objectives, several practical limitations were identified during development:

- **Online Payment Gateway:** The current version does not integrate an automated online payment processing gateway (such as PayHere or Stripe) for paid event tickets; ticket fee settlements are currently verified manually.
- **Offline PWA Caching:** The system relies on active internet connectivity to communicate with the REST API and MySQL server, without full offline Progressive Web App service worker caching.
- **Third-Party Messaging:** Direct automated SMS ticket dispatch is not currently configured, relying instead on browser-generated printable passes and on-screen QR codes.

6.4 Recommendations for Future Enhancements

To further expand the operational capabilities of the Rotaract Club of NIBM Kandy Management System, the following future enhancements are recommended:

1. **Rotary International API Integration:** Integrate direct API webhooks with Rotary District 3220 and Rotary International databases to synchronize annual club citation statistics automatically.
2. **Payment Gateway Integration:** Incorporate a secure payment gateway for automated fundraising and ticket sales processing.
3. **Native Mobile Application:** Develop cross-platform mobile companion apps using React Native or Flutter to provide push notifications and offline QR scanning for event marshals.
4. **Automated Email/SMS Dispatch:** Integrate transactional email services (Nodemailer / SendGrid) and SMS gateways (Twilio / Dialog Ideamart) for automated pass notifications.

6.5 Chapter Summary

In conclusion, the Rotaract Club of NIBM Kandy Management System represents a significant technological advancement over existing manual practices. The successful delivery of the system demonstrates the practical application of software engineering principles, agile development

methodologies, full-stack MVC architecture, and relational database management to deliver tangible value to student-led community organizations.

REFERENCES

- Bass, L., Clements, P. and Kazman, R. (2021) *Software Architecture in Practice*. 4th edn. Boston: Addison-Wesley Professional.
- Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R.C., Mellor, S., Schwaber, K., Sutherland, J. and Thomas, D. (2001) *Manifesto for Agile Software Development*. Available at: <https://agilemanifesto.org/> (Accessed: 15 August 2025).
- Connolly, T.M. and Begg, C.E. (2015) *Database Systems: A Practical Approach to Design, Implementation, and Management*. 6th edn. Boston: Pearson.
- Elmasri, R. and Navathe, S.B. (2015) *Fundamentals of Database Systems*. 7th edn. Hoboken: Pearson.
- Fielding, R.T. (2000) *Architectural Styles and the Design of Network-based Software Architectures*. Doctoral dissertation. University of California, Irvine.
- Gamma, E., Helm, R., Johnson, R. and Vlissides, J. (1994) *Design Patterns: Elements of Reusable Object-Oriented Software*. Reading: Addison-Wesley.
- MySQL AB (2024) *MySQL 8.0 Reference Manual*. Oracle Corporation. Available at: <https://dev.mysql.com/doc/refman/8.0/en/> (Accessed: 18 August 2025).
- Node.js Foundation (2024) *Node.js v20 LTS Documentation*. OpenJS Foundation. Available at: <https://nodejs.org/docs/latest-v20.x/api/> (Accessed: 16 August 2025).
- Pressman, R.S. and Maxim, B.R. (2020) *Software Engineering: A Practitioner's Approach*. 9th edn. New York: McGraw-Hill Education.
- React Documentation (2024) *React: The library for web and native user interfaces*. Meta Open Source. Available at: <https://react.dev/> (Accessed: 12 August 2025).

Rotary International (2023) *Rotaract Handbook: A Guide for Clubs and Sponsors*. Evanston: Rotary International Publishing.

Sommerville, I. (2016) *Software Engineering*. 10th edn. Harlow: Pearson Education Limited.

Tailwind Labs Inc. (2024) *Tailwind CSS Documentation: A utility-first CSS framework*. Available at: <https://tailwindcss.com/docs> (Accessed: 14 August 2025).